KATIE **GROEGER**

(607) 368-7684
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SUMMARY

Highly motivated Associate Scientist with 9 years of industry experience in molecular biology. Skilled in designing high-throughput workflows and automation processes to drive innovation. Eager to leverage and expand my expertise in a research and development role focused on cutting-edge biotechnological solutions.

EDUCATION

Boston University
M.S. Computer Science and AI
Expected May 2028

Northeastern University
M.S. Bioengineering: Cell & Tissue Engineering
4.0 GPA
May 2022

Rensselaer Polytechnic Institute
B.S. Chemical Engineering
May 2016

SKILLS

Molecular Biology:
- DNA/RNA/Protein Purification
- RNA Production
- LNP Formulation
- Antibody Conjugation
- Cell Culture
- Tissue Culture
- Cloning

Data Analysis:
- Python
- Excel
- JMP
- GraphPad

Automation:
- Hamilton Venus method writing
- High Throughput Production

Technical Writing:
- SOP & Technical Document Writing

CERTIFICATIONS

- MIT edX: Introduction to Computer Science and Programming Using Python
- Hamilton Venus Software Training I
- Six Sigma Yellow Belt Training

INTERESTS

- Scuba Diving
- Mediocre Skier
- Avid Reader

PROFESSIONAL EXPERIENCE

CRISPR Therapeutics, Boston, MA **Aug 2023 - Present**

Senior Research Associate | Aug 2023-Present
- Optimized lipid nanoparticle (LNP) formulations to improve encapsulation, organ targeting, and therapeutic efficacy
- Prepared LNPs for in vitro and in vivo studies to validate LNP performance and refine formulation strategies
- Collaborated with cross-functional teams, presenting findings and recommendations to internal stakeholders and external partners

GreenLight Biosciences Inc., Lexington, MA **Jan 2017 - July 2023**

Associate Scientist | Jan 2022-July 2023
- Human Health mRNA Upstream Production HTP platform development:
  - Produced, analyzed, and delivered mRNA from pDNA libraries
  - Designed and executed mRNA titer optimization experiments
  - Created new Venus methods for Hamilton STAR liquid handling robots to complete assay preparations
  - Updated Venus methods created for Plant Health HTP to be effective for Human Health mRNA production
- Plant Health HTP Platform development and production optimization:
  - Developed and tested scale-up experiments to increase dsRNA titer 20x
  - Designed and performed high throughput RNA experiments on Hamilton STAR liquid handling robots for optimization of reagent cost, RNA titer, product quality, and time

Senior Research Associate | Jan 2021 - Jan 2022
- HTP Platform Development and RNA production optimization for Plant Health and Human Health programs
- Led high-throughput dsRNA production on Hamilton STAR liquid handlers for a plant health platform; trained new team members and served as a subject matter expert on workflow, data analysis, and troubleshooting

Research Associate III, Pathway Engineering | Feb 2019 - Jan 2021
- Designed, performed, and analyzed series of enzyme assays used to develop a scale-up method
- Generated and confirmed COVID-19 DNA via Giga prep for mRNA vaccine development
- Designed and performed in vitro transcription optimization experiments for the development of mRNA vaccines

Research Associate II, Pathway Engineering | May 2018 - Feb 2019
- Transformed bacterial cells to create cryo-stocks using electroporation cuvettes and plate-based methods, and performed quality control via LabChip, SDS PAGE, agarose gels and BCA
- Purified lysate via AKTA FPLC and spin column purification methods

Research Associate, Process Development | Jan 2017 - May 2018
- Generated bacterial biomass on the 1-liter scale to be used in dsRNA generation for a plant and animal health platform
- Initiated fermentation experiments performed in 70 mL bioreactors